

True or false? (unless otherwise specified)

1. If $A \in \mathbf{R}^{n \times n}$ is singular (non-invertible), then for any vector $\mathbf{b} \in \mathbf{R}^n$, there exist infinitely many solutions to the linear system

$$A\mathbf{x} = \mathbf{b}.$$

2. Suppose that $(\mathbf{x}_i)_{i \in \mathbf{N}}$ is a sequence in $\mathbf{R}^n$ and that $\mathbf{x}_* \in \mathbf{R}^n$. Then we have the equivalence

$$\lim_{i \rightarrow \infty} \|\mathbf{x}_i - \mathbf{x}_*\|_\infty = 0 \quad \Leftrightarrow \quad \lim_{i \rightarrow \infty} \|\mathbf{x}_i - \mathbf{x}_*\|_1 = 0.$$

3. For a vector norm $\|\cdot\|$ on $\mathbf{R}^n$, the *subordinate* or *induced* matrix norm is defined by

$$\|M\| := \max\{\|M\mathbf{x}\| : \|\mathbf{x}\| \leq 1\}.$$

Then it holds that $\|AB\| \leq \|A\|\|B\|$ for all $A, B \in \mathbf{R}^{n \times n}$.

4. Suppose that $A \in \mathbf{R}^{n \times n}$ is an invertible matrix and consider a splitting $A = M - N$, with M an invertible matrix. Suppose that $\mathbf{b} \in \mathbf{R}^n$ is given and $\mathbf{x}^{(0)} = \mathbf{0} \in \mathbf{R}^n$. Consider the following iterative method:

$$M\mathbf{x}^{(k+1)} = N\mathbf{x}^{(k)} + \mathbf{b}, \quad (1)$$

Denote by $\mathbf{x}_*$ the exact solution to the linear system $A\mathbf{x} = \mathbf{b}$, and recall that, as we proved in class, the error $\mathbf{e}^{(k)} := \mathbf{x}^{(k)} - \mathbf{x}_*$ satisfies the equation

$$\mathbf{e}^{(k)} = (M^{-1}N)^k \mathbf{e}^{(0)}. \quad (2)$$

Then the error satisfies the inequality

$$\|\mathbf{e}^{(k)}\|_\infty \leq \|M^{-1}N\|_\infty^k \|\mathbf{e}^{(0)}\|_\infty.$$

5. For the iterative method (1), the approximation $\mathbf{x}^{(k)}$ converges as $k \rightarrow \infty$ to the exact solution $\mathbf{x}_*$ if and only if the following inequality is satisfied: $\|M^{-1}N\|_\infty < 1$.
6. The Jacobi method is an iterative method of the form (1) for the splitting $M = D$ and $N = -L - U$, where matrix D is the diagonal part of A , and L, U are the strictly lower and upper triangular parts, respectively. Then, for a general matrix A , each iteration of this method requires $\mathcal{O}(n)$ floating point operations.
7. Assume that $\mathbf{b} \in \mathbf{R}^n$ and that $A \in \mathbf{R}^{n \times n}$ is symmetric and positive definite. Then a vector $\mathbf{x}_*$ satisfies the equation $A\mathbf{x}_* = \mathbf{b}$ if and only if

$$f(\mathbf{x}_*) = \min_{\mathbf{x} \in \mathbf{R}^n} f(\mathbf{x}), \quad \text{where } f(\mathbf{x}) := \frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{b}^T \mathbf{x}. \quad (3)$$

8. Assume that $\mathbf{b} \in \mathbf{R}^n$ and that $A \in \mathbf{R}^{n \times n}$ is symmetric and positive definite. Assume additionally that the vectors $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ are A -conjugate, meaning that $\mathbf{e}_i^T A \mathbf{e}_j = 0$ if $i \neq j$, and denote by $\mathbf{x}_*$ the exact solution to the linear system $A\mathbf{x} = \mathbf{b}$. Then it holds that

$$\mathbf{x}_* = \frac{\mathbf{e}_1^T \mathbf{b}}{\mathbf{e}_1^T A \mathbf{e}_1} \mathbf{e}_1 + \dots + \frac{\mathbf{e}_n^T \mathbf{b}}{\mathbf{e}_n^T A \mathbf{e}_n} \mathbf{e}_n.$$

9. Suppose that $A \in \mathbf{R}^{2 \times 2}$ is symmetric, with a positive eigenvalue and a negative eigenvalue. Then the function f defined in (3) does not have a minimizer, and furthermore

$$\inf_{\mathbf{x} \in \mathbf{R}^n} f(\mathbf{x}) = -\infty.$$

10. The following code implements an iterative method for solving $A\mathbf{x} = \mathbf{b}$. (Note that the matrix A here is symmetric and positive definite.) In this case the method does not converge; the `while` loop never terminates.

```
A = [4 1 0; 1 4 1; 0 1 4]
b = [1.; 2.; 3.]
x = [0.; 0.; 0.]
r = A*x - b
while sqrt(r'r) > 1e-12 * sqrt(b'b)
    r = A*x - b
    w = r'r / (r'A*r)
    x -= w * r
end
```

11. **Bonus:** The following code implements a basic iterative method based on a splitting. What is the name of the method implemented? Also give the explicit expressions of M and N .

```
A = [4 1 0; 1 4 1; 0 1 4]
b = [1.; 2.; 3.]
x = [0.; 0.; 0.]
for k in 1:100
    x[1] = (b[1] - A[1, 2] * x[2] - A[1, 3] * x[3]) / A[1, 1]
    x[2] = (b[2] - A[2, 1] * x[1] - A[2, 3] * x[3]) / A[2, 2]
    x[3] = (b[3] - A[3, 1] * x[1] - A[3, 2] * x[2]) / A[3, 3]
end
```

Your answer:

12. **Bonus:** Prove the equation (2) satisfied by the error for the basic iterative method based on a splitting.

Your answer:

Solutions

1. **False.** A singular matrix $\mathbf{A}$ does not necessarily yield infinitely many solutions. The system $\mathbf{Ax} = \mathbf{b}$ has solutions only when $\mathbf{b} \in \text{range}(\mathbf{A})$; otherwise it has no solution. Example: if $\mathbf{A} = \mathbf{0}$, then no solution exists unless $\mathbf{b} = \mathbf{0}$.

2. **True.** In $\mathbb{R}^n$ all norms are equivalent. In particular,

$$\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_1 \leq n\|\mathbf{v}\|_\infty,$$

so $\|\mathbf{x}_i - \mathbf{x}_*\|_\infty \rightarrow 0$ iff $\|\mathbf{x}_i - \mathbf{x}_*\|_1 \rightarrow 0$.

3. **True.** For any $\mathbf{x}$ with $\|\mathbf{x}\| \leq 1$,

$$\|\mathbf{ABx}\| = \|\mathbf{A(Bx)}\| \leq \|\mathbf{A}\| \|\mathbf{Bx}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|,$$

and maximizing over $\|\mathbf{x}\| \leq 1$ gives $\|\mathbf{AB}\| \leq \|\mathbf{A}\| \|\mathbf{B}\|$.

4. **True.** Using the error formula $\mathbf{e}^{(k)} = (\mathbf{M}^{-1}\mathbf{N})^k \mathbf{e}^{(0)}$ and submultiplicativity,

$$\|\mathbf{e}^{(k)}\|_\infty \leq \|(\mathbf{M}^{-1}\mathbf{N})^k\|_\infty \|\mathbf{e}^{(0)}\|_\infty \leq \|\mathbf{M}^{-1}\mathbf{N}\|_\infty^k \|\mathbf{e}^{(0)}\|_\infty.$$

5. **False.** The condition $\|\mathbf{M}^{-1}\mathbf{N}\|_\infty < 1$ is *sufficient* for convergence but not *necessary*. The correct necessary and sufficient condition is $\rho(\mathbf{M}^{-1}\mathbf{N}) < 1$, where $\rho(\cdot)$ denotes the spectral radius.

6. **False.** For a general dense matrix $\mathbf{A}$, one Jacobi iteration requires $\mathcal{O}(n^2)$ operations: each of the n components requires summing approximately n terms. The cost is $\mathcal{O}(n)$ only for banded or sparse matrices.

7. **True.** Since $\mathbf{A}$ is symmetric positive definite, $f(\mathbf{x})$ is strictly convex and

$$\nabla f(\mathbf{x}) = \mathbf{Ax} - \mathbf{b}.$$

Thus f is minimized exactly at points satisfying $\mathbf{Ax}_* = \mathbf{b}$.

8. **True.** Since the $\mathbf{e}_i$ form an $\mathbf{A}$ -conjugate basis, write $\mathbf{x}_* = \sum_{i=1}^n \alpha_i \mathbf{e}_i$. Taking $\mathbf{e}_j^\top \mathbf{A}(\cdot)$ gives $\mathbf{e}_j^\top \mathbf{b} = \alpha_j \mathbf{e}_j^\top \mathbf{A} \mathbf{e}_j$, hence

$$\mathbf{x}_* = \sum_{i=1}^n \frac{\mathbf{e}_i^\top \mathbf{b}}{\mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i} \mathbf{e}_i.$$

9. **True.** If $\mathbf{A}$ has a positive and a negative eigenvalue, then along the eigenvector $\mathbf{v}$ associated with the negative eigenvalue $\lambda < 0$,

$$f(t\mathbf{v}) = \frac{1}{2}\lambda t^2 - t\mathbf{v}^\top \mathbf{b} \rightarrow -\infty \quad \text{as } |t| \rightarrow \infty.$$

Thus f has no minimizer and $\inf f = -\infty$.

10. **False (the claim of nonconvergence is false).** The code implements the steepest-descent method with exact line search: for $\mathbf{A}$ symmetric positive definite, this method always converges. Thus the loop should terminate (modulo tolerance issues), not run forever.

Bonus 1. The method is **Gauss–Seidel**. With the splitting $\mathbf{A} = \mathbf{D} + \mathbf{L} + \mathbf{U}$, the Gauss–Seidel iteration uses

$$\mathbf{M} = \mathbf{D} + \mathbf{L}, \quad \mathbf{N} = -\mathbf{U},$$

so that

$$(\mathbf{D} + \mathbf{L})\mathbf{x}^{(k+1)} = -\mathbf{U}\mathbf{x}^{(k)} + \mathbf{b}.$$

Bonus 2. Starting from

$$\mathbf{M}\mathbf{x}^{(k+1)} = \mathbf{N}\mathbf{x}^{(k)} + \mathbf{b},$$

and using the fact that $\mathbf{M}\mathbf{x}_* = \mathbf{N}\mathbf{x}_* + \mathbf{b}$ (since $\mathbf{A} = \mathbf{M} - \mathbf{N}$ and $\mathbf{A}\mathbf{x}_* = \mathbf{b}$), subtract the equations to obtain

$$\mathbf{M}(\mathbf{x}^{(k+1)} - \mathbf{x}_*) = \mathbf{N}(\mathbf{x}^{(k)} - \mathbf{x}_*).$$

Hence

$$\mathbf{e}^{(k+1)} = \mathbf{M}^{-1}\mathbf{N}\mathbf{e}^{(k)}.$$

Iterating yields

$$\mathbf{e}^{(k)} = (\mathbf{M}^{-1}\mathbf{N})^k \mathbf{e}^{(0)}.$$